

Dr. Purba Das

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Professional activity

Sep 23-Present **Lecturer (Assistant Professor)**, Department of Mathematics, King's College London (KCL).
Aug 22 - Aug 23 **Byrne Research Assistant Professor**, Department of Mathematics, University of Michigan, Ann Arbor.

Academic history

Oct 2018–May 2022 **DPhil in Mathematics**, University of Oxford
PhD thesis, Roughness properties of paths and signals, **Advisor: Prof. Rama Cont**.
2016–2018 **Masters**, Applications of Mathematics, Chennai Mathematical Institute (CMI), India.
Master's thesis, On completely mixed stochastic games, **Advisor: Prof. T Parthasarathy**.
2013–2016 **Undergraduate**, Honours in Mathematics and Computer Science. Chennai Mathematical Institute (CMI), India.

Publications and preprints

PREPRINTS/ UNDER SUBMISSION

Quadratic and p-th variation of random signed Takagi–Landsberg bridges

PURBA DAS, ALEXANDER SCHIED

arXiv preprint

2026

Banach spaces of continuous paths with finite p-th variation

PURBA DAS, DONGHAN KIM, FANG RUI LIM

arXiv preprint

2026

Gamma Hedging without Rough Paths

JOHN ARMSTRONG, PURBA DAS

arXiv preprint

2026

Noise estimation of SDE from a single data trajectory

MUNAWAR ALI, PURBA DAS, QI FENG, LIYAO GAO, GUANG LIN

arXiv preprint

2025

A rough path approach to pathwise stochastic integration à la Föllmer

PURBA DAS, ANNA P. KWOSSEK, DAVID J. PRÖMEL

arXiv preprint

2025

Pathwise Itô isometry for scaled quadratic variation

SUPRIO BHAR, PURBA DAS, BARUN SARKAR

arXiv preprint

2025

X marks the bank: digitalisation and depositor runs

ARTUR KOTLICKI, PURBA DAS AND VIRGINIA PUGLIESE

SSRN Preprint

2025

PUBLISHED ¹

¹All papers are available open access on arXiv.

Local times of deterministic paths and self-similar processes with stationary increments as normalized numbers of interval crossings.

WITOLD BEDNORZ, **PURBA DAS** AND RAFAŁ ŁOCHOWSKI

2026

Ann. Inst. H. Poincaré Probab. Statist. 62(1): 523-547 (February 2026). DOI: 10.1214/24-AIHP1519

Level crossings of fractional Brownian motions

PURBA DAS AND RAFAŁ ŁOCHOWSKI AND TOYOMU MATSUDA AND NICOLAS PERKOWSKI

2025

Annals of Probability, 2025, Vol. 53, No. 6, 2354–2393

On isomorphism of the space of continuous functions with finite p -th variation along a partition sequence

PURBA DAS AND DONGHAN KIM

2025

Journal de Mathématiques Pures et Appliquées, Volume 203, November 2025, 103753

Hölder regularity and roughness: construction and examples

ERHAN BAYRAKTAR AND **PURBA DAS** AND DONGHAN KIM

2025

Bernoulli 31(2): 1084–1113 (May 2025). DOI: 10.3150/24-BEJ1761

Rough volatility: fact or artefact?

RAMA CONT AND **PURBA DAS**

2024

Sankhya B, Volume 86, pages 191–223 (2024)

Pathwise quadratic variation and quadratic roughness

RAMA CONT AND **PURBA DAS**

2023

Bernoulli 29(1): 496–522 (February 2023). DOI: 10.3150/22-BEJ1466

Quadratic variation along refining partitions: Constructions and examples

RAMA CONT AND **PURBA DAS**

2022

Journal of Mathematical Analysis and Applications, Volume 512, Issue 2, 15 August 2022, 126173

On Completely Mixed Stochastic Games

PURBA DAS, T. PARTHASARATHY, G. RAVINDRAN.

2022

Operations Research Forum volume 3, Article number: 57 (2022)

Understanding Sea Ice Melting via Functional Data Analysis

PURBA DAS, ANANYA LAHIRI, SOURISH DAS

2018

Current Science, Volume 115, Issue 5, Pages 920–929.

Scholarships and Grants

June 26 – Feb 26	Heilbronn International Visitors Scheme , a £6.1k grant for hosting Alexander Schied
Sep 2026 - Mar 2027	LMS, Collaborations with Developing Countries - Scheme 5 , a £3k grant for hosting visitor
Sep 2025-28	PhD student funding , Funding to support a PhD student, covering tuition fees at KCL as well as a monthly stipend.
Dec 2024- May 2025	KCL-NMES Industry engagement grant , a £5k grant jointly with Bank of England
Jan-July 2024	IIT Madras and KCL joint seed fund grant , A £10K grant to initiate a research collaboration
2018-2022	Mathematical Institute Studentship at University of Oxford , A full waiver of the tuition fees and a monthly fellowship towards living costs
2016-2018	Postgraduate studentship at Chennai Mathematical Institute, India , A full waiver of the tuition fees and a monthly fellowship towards living costs
2013-2016	Undergraduate studentship at Chennai Mathematical Institute, India , A full waiver of the tuition fees and a monthly fellowship towards living costs

Supervision

PHD SUPERVISION

2024-Present	Rohan Hobbs , Second supervisor;	Primary supervisor – John Armstrong
2025-Present	Martin Dattge , Primary supervisor;	Second supervisor – David Prömel

OTHER SUPERVISION

2026 summer	MSc thesis/ project supervision , total 6 students, including students from King's College London and the University of Edinburgh.
2025 summer	MSc thesis/ project supervision , total 8 students, including students from King's College London and the University of Edinburgh.
2024 summer	MSc thesis/ project supervision , total 9 students from King's College London
2025 summer	BSc thesis/ project supervision , 1 student from University of Michigan.
	Personal tutor , I have been acting as a personal tutor for approx 30 students each year since
Sep 23- Present	September 2023- These students are a combination of undergrad and Master at King's College London

Talks and poster presentations

Feb 2027	Invited talk , Actuarial Science and Mathematical Finance seminar at Amsterdam	Amsterdam
Dec 2026	Invited talk , SLAM 2026 link	Liège
Nov 2026	Invited talk , Berlin Probability colloquium	Berlin
Oct 2026	Invited talk , Mathematical finance and stochastic analysis seminar at York	York
June 2026	Invited talk , Department of Statistics and Operations Research, University of Vienna	Austria
May 2026	Invited talk , The rough path interest group, The Alan Turing Institute	London/online
May 2026	Invited talk , Ahoka University, Department of Mathematics	India
May 2026	Invited talk , Ahoka University, Safexpress Centre for Data, Learning and Decision Sciences	India
Mar 2026	Invited talk , Mathematical Finance seminars, University of Birmingham	Birmingham
Dec 2025	Invited talk , Oxford CDT workshop–Mathematics of random system	Oxford
Dec 2025	Invited talk , Mathematical Finance and Insurance Mathematics Seminar at LMU Munich	Munich
Nov 2025	Invited talk , 12th International Conference of Image Processing, Wavelet and Applications on Real World Problems	Online
Sep 2025	Invited talk , Workshop: Young Researchers in Stochastic Analysis and Stochastic Geometric Analysis	Lausanne
May 2025	Invited talk , Probability Seminar, University of Leeds	Leeds
Nov 2024	Invited talk , QuantMinds International 2024	London
Sep 2024	Invited talk , Mathematical insights from markets, control, and learning	France
June 2024	Invited talk , Mathematics Seminar, Chennai Mathematical Institute	India
June 2024	Invited talk , International Conference on Stochastic Calculus and Applications to Finance	IIT Madras, India
May 2024	Invited talk , KCL Analysis seminar, King's College London	London
Apr 2024	Invited talk , Probability and Applications seminar, Queen Mary	London
Dec 2023	Invited talk , Quant Summit Europe 2023	London
Nov 2023	Invited talk , QuantMinds International	London
Oct 2023	Invited talk , London Financial Mathematics seminar	UCL, London
Oct 2023	Invited talk , Stochastic Finance Seminar, University of Warwick	Warwick
Oct 2023	Invited talk , Probability seminar, University of Manchester	Manchester
Oct 2023	Invited talk , Financial Mathematics seminar, King's College London	London

Sep 2023	Invited talk , London-Paris Bachelier workshop 2023	London
Sep 2023	Invited talk , Financial mathematics seminar, Florida State University	Online
Feb 2023	Invited talk , IFAM, University of Liverpool	Online
Sep 2022	Contributed talk , Financial Mathematics Seminar, University of Michigan	Ann Arbor, USA
June 2022	Contributed talk , Oxford-ETH Workshop on Mathematical & Computational Finance	Zurich
June 2022	Contributed talk , 11th World Congress of the Bachelier Finance Society	Online
May, 2022	Contributed talk , 15th Berlin-Oxford Young Researchers Meeting on Applied Stochastic Analysis	Berlin, Germany
May, 2022	Invited talk , Dynamical Systems/Stochastics Seminar, Free University, Berlin	Berlin, Germany
April, 2022	Invited talk , IRTG Graduate Student Seminar, Technische Universität Berlin	Berlin, Germany
Apr, 2022	Invited talk , Mathematical Finance Seminar, Imperial College London	London, UK
Jan, 2022	Invited talk , Mathematical Finance Seminar, University of Michigan	Ann Arbor, USA
Oct, 2021	Contributed talk , Mathematical and Computational Finance Internal Seminar	Oxford, UK
Oct, 2021	Contributed talk , Junior female researchers in probability	Berlin, Germany
Aug, 2021	Contributed talk , 6th Berlin Workshop for Young Researchers on Mathematical Finance	Online
July, 2021	Contributed talk , Bernoulli-IMS 10th World Congress in Probability and Statistics	Online
July, 2021	Contributed talk , Young Researcher's Session, Statistical Methods in Finance 2021	Online
Jan 2020	Contributed talk , Bachelier Colloquiums, Mathematical Finance and stochastic calculus	Métabief, France
July 2019	Poster presentation , LMS Research School on Probability	Bath, UK
March 2019	Contributed Talk , Oxford-ETH Workshop on Mathematical Finance.	Oxford, UK
April 2018	Invited Talk , Internal Seminar at Indian Statistical Institute.	Chennai, India
April 2018	Research seminar talk , Chennai Mathematical Institute (CMI), India.	Chennai, India
March 2018	Invited talk , Research seminar, Indian Institute of Technology, India.	Chennai, India
Dec, 2017	Poster presentation , International conference on statistics.	Hyderabad, India
Sep 2016	Contributed talk , Workshop on Linear Complementarity and Generalisations.	ISI Chennai, India

Teaching

PROFESSIONAL RECOGNITION

April 2026	Fellow of the Higher Education Academy (FHEA), UK , a recognised certification in university-level teaching and student support ((Link to Advance HE).
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PAST AND PRESENT TEACHING

Spring 2027	Lecturer: 6CCM338A - Mathematical Finance II , Department of Mathematics, King's College London.
Spring 2027	Lecturer: 7CCMFM04 - Stochastic Analysis , Department of Mathematics, King's College London.
Spring 2026	Lecturer: 6CCM338A - Mathematical Finance II , Department of Mathematics, King's College London.
Spring 2025	Lecturer: 6CCM338A - Mathematical Finance II , Department of Mathematics, King's College London.
Spring 2024	Lecturer: 6CCM338A - Mathematical Finance II , Department of Mathematics, King's College London.
Winter 2023	Lecturer: MATH 526- Discrete Stochastic Process , Department of Mathematics, University of Michigan.
Fall 2022	Lecturer: MATH 423- Mathematics of Finance , Department of Mathematics, University of Michigan.
HT 2022	Teaching Assistance: Integration , Queen's College, University of Oxford.
MT 2021	Tutor: Probability, measure, and martingale , Mathematical Institute, University of Oxford.
TT 2021	Tutor: Continuous Martingales and Stochastic Calculus , Mansfield College, University of Oxford.
HT 2021	Tutor: Continuous Martingale and Stochastic Calculus , Mathematical Institute, University of Oxford.
HT 2021	Tutor: Optimisation , Mathematical Institute, University of Oxford.
HT 2021	Tutor: Stochastic Calculus , Mathematical Institute, University of Oxford.
MT 2020	Tutor: Probability- Measure and Martingales , Mathematical Institute, University of Oxford.

HT 2020	Teaching Assistance: Optimisation , Mathematical Institute, University of Oxford.
HT 2020	Teaching Assistance: Continuous Martingale and Stochastic Calculus , Mathematical Institute, University of Oxford.
MT 2019	Teaching Assistance: Stochastic Differential Equations , Mathematical Institute, University of Oxford.
MT 2019	Teaching Assistance: Probability, Measure and Martingales , Mathematical Institute, University of Oxford.
HT 2019	Teaching Assistance: Continuous Optimisations , Mathematical Institute, University of Oxford.
2018	Teaching Assistance: Game Theory , Chennai Mathematical Institute.
2017	Teaching Assistance: Regression and Classification , Chennai Mathematical Institute, India.
2016	Teaching Assistance: Optimisation techniques , Chennai Mathematical Institute, India.
2016	Teaching Assistance: Financial Risk Management , Chennai Mathematical Institute, India.

Conference and seminar organisations

Sep 2026	10th London–Paris Bachelier Workshop on Mathematical Finance , Co-Organiser	<i>Paris</i>
July 2026	Quant Finance Day , Co-Organiser with Luca Galimberti (KCL) and Nimit Rana (York)	<i>York</i>
2024-Present	London Mathematical Finance Seminar Series , A biweekly seminar series jointly organised by KCL, UCL, LSE, IC	<i>London</i>
2024-Present	King’s College London Financial Mathematics Seminars , A biweekly seminar series	<i>KCL, London</i>
Nov 2025	9th London–Paris Bachelier Workshop on Mathematical Finance , Co-Organiser	<i>LSE, London</i>
April 2025	Conference on Liquidity Stresses and Financial Stability , Co-Organiser with Artur Kotlicki (Bank of England)	<i>BoE, London</i>
June 2024	International Conference on Stochastic Calculus and Applications to Finance- with a focus towards Functional Itô calculus and Stochastic PDEs in distribution space , Co-organised with Barun Sarkar (IIT Madras, India) and Suprio Bhar (IIT Kanpur, India)	<i>IIT Madras, India</i>

Miscellaneous (Including research visit)

Tentative Nov 2026	Research visit at Liège University, Hosted by Céline Esser
June 2026	Research visit at University of Vienna, Hosted by Christa Cuchiero
2025-Present	Mentor for Women in Applied Mathematics Mentoring Scheme (WAMMS)
July 2022	Research visit at Warsaw School of Economics, Hosted by Rafał Łochowski
Apr-May 2022	Research visit at Freie Universität Berlin, Hosted by Nicholas Perkowski
Jun 2021	Best student paper at Statistical Methods in Finance 2021.
2020-2022	Organising committee for Mathematrix
2020-2022	Organising committee for Women’s coffee
March 2018	First Prize for the best poster at ‘Tata Consultancy Services (TCS) Poster Competition’.
Dec 2017	Honourable mention for the poster presentation at International Conference on Statistics.
2016	Organising committee for Annual Fest in Chennai Mathematical Institute